

Synopsis: Hysterical Response Halo and Galactic Scaling

What the theorems say (without the proofs)

Andrew Bruce Boyd

September 2026

Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

This is a short guide to the main claims of the galactic modelling paper. It builds on the Hysterical Response of the earlier response-theory paper. Each section says *why* the claim matters, then states the result. The proofs live in the full paper.

1 From Hysterical Response to a galactic halo

The earlier response-theory paper isolates the Hysterical Response: the residual force of an already-existing interaction under exponentially stable open-system dynamics,

$$F_H = -\text{Tr}(F \mathcal{L}^{-1} \mathcal{S}), \quad (1)$$

equivalently $\delta\rho_{ss} = -\mathcal{L}^{-1}\mathcal{S}$. No new force carrier is introduced: F_H is a component of the response of the ordinary force F itself. That paper stops at channel geometry, neutrality, instability, and infrared equivalence. It does *not* build galaxies.

The present paper takes that gravitational Hysterical Response as the force and continues it into an explicit spatial halo. A ballistic Wigner representation of the residual response (production source S , loss rate Γ) is phase-space bookkeeping for how the residue is organized in space—not a claim of a new messenger particle. Here S is the ballistic image of the companion paper’s residual drive $\mathcal{S}$. The retarded characteristic solution of $v \cdot \nabla f = S - \Gamma f$ is $f(x, v) = \int_0^\infty e^{-\Gamma t} S(x - tv, v) dt$ (Lean-checked under standard regularity). The total production rate $Q = \int J$ becomes the amplitude that later sets galactic scales. If that ballistic J packages Paper 1’s unentangled-incoming drive, then $Q = \alpha J_{\text{in}}^{\text{tot}}$; when baryonic accretion supplies those arrivals, $Q = \alpha \dot{N}_{\text{acc}}$ (conversion α remains model-dependent—not the unit identity $Q = \dot{M}_{\text{b}}$). “Hysterical Response Halo” means that extended residual-response density (and the acceleration it produces under the infrared closure)—ordinary gravity’s leftover, not a new force law invented here from scratch.

2 Two galactic problems

Observed rotation curves stay roughly flat where Newtonian baryons would already be Keplerian, and the asymptotic speed tracks baryonic mass as $M_b \propto v_f^4$ (BTFR). Those are different requirements: one is a radial profile $g_{\text{resp}} \propto 1/r$; the other is a mass normalization. The full paper separates them.

3 Ballistic response halo

Under an infrared ballistic realization—local velocity isotropy plus a single characteristic speed v —residual-response production takes the monoenergetic shell form $S = J \delta(|u| - v)/(4\pi v^2)$. For a localized, centered spatial profile J , the stationary density then has a universal leading far field.

Theorem 3.1 (Ballistic response halo). *Outside the source and well inside the propagation length,*

$$n = \frac{Q}{4\pi v r^2} + O(r^{-4}).$$

The leading halo is spherical; source shape enters only at higher order.

4 Why $g_{\text{resp}} = CQ/r$

A density $n \propto Q/r^2$ is not yet a force; the force is already the Hysterical Response of gravity. The primary modelling closure is the linear Poisson law for the associated response potential

$$\nabla^2 \Phi_{\text{resp}} = \kappa n. \quad (2)$$

Together with the ballistic halo and $\nabla^2 \ln r \propto r^{-2}$, this yields

$$g_{\text{resp}} = \frac{CQ}{r} \quad (3)$$

for a coupling $C = \kappa/(4\pi v) > 0$. Separate-field or unified-field language may be used as brief physical readings of that same infrared law; they do not independently postulate CQ/r .

Theorem 4.1 (Flat response rotation). *Circular balance $v_c^2/r = CQ/r$ then implies $v_c^2 = CQ$, hence an asymptotic response-dominated speed $v_f^2 = CQ$.*

Flat curves are therefore the radial consequence of the Hysterical Response halo plus the Poisson closure; they do not by themselves fix how Q scales with baryonic mass.

5 Capacity attractor vs formation conjecture

The ballistic spatial model fixes the shape of $n(r)$, not the pair (Q, M_b) . Introduce a capacity coefficient $A > 0$ and filling fraction $y = A^2 M_b / Q^2$ (equivalently the capacity inequality $M_b \leq Q^2 / A^2$ when $y \leq 1$). Response-limited assembly is modelled by a logistic ODE for y with stable fixed point $y_* \in (0, 1]$.

Corollary 5.1 (Square-root attractor). *Given that capacity inequality / filling setup and the stable assembly dynamics, at y_* one has*

$$Q^2 = \frac{A^2}{y_*} M_b, \quad (4)$$

so $Q \propto \sqrt{M_b}$. This attractor is a theorem/corollary of those hypotheses—not a conjecture.

Combined with $v_f^2 = CQ$,

$$v_f^4 = C^2 Q^2 \propto M_b. \quad (5)$$

Corollary 5.2 (BTFR exponent). *On that attractor, $v_f^4 \propto M_b$. The zero point remains encoded in C , A , and y_* ; those microscopic values are not derived here.*

Conjecture 1 (Formation / retention capacity). What is conjectural is the physical selection rule: galactic formation and mature retention require

$$Q \gtrsim O(\sqrt{M_b}).$$

That requirement is not proved from the ballistic spatial model (the attractor corollary places mature systems on $Q \sim \sqrt{M_b}$ once capacity filling is assumed).

6 Late-time size–mass law

With isotropic accretion ($j \propto M_b^{2/3}$) and size proxy $R \sim j/v_f$, the BTFR velocity scaling implies a definite size–mass exponent. The isotropic- j input is physically motivated by late-time feeding from a homogeneous intergalactic dust/gas reservoir (no preferred inflow direction); filamentary anisotropy would change the exponent.

Theorem 6.1 (Late-time size–mass). *On the mature attractor $v_f^4 = KM_b$,*

$$R \propto M_b^{5/12}.$$

Observed late-type slopes typically range from ~ 0.22 (CANDELS late-types) through ~ 0.39 (high-mass SDSS late-types) to ~ 0.385 (baryonic disc scale lengths). The prediction $5/12 \approx 0.42$ sits near the steeper end of that range.

7 Asymptotic oblateness

The first nonspherical correction for a planar axisymmetric source is a transport quadrupole. Under the same primary Poisson closure it produces an oblate potential whose effective flattening satisfies

$$q_{\text{eff}}^{-2}(R) = 1 + \frac{s}{R^2} + O(R^{-4}),$$

with $s = \langle R'^2 \rangle$ (and $s = 6R_d^2$ for an exponential disc). Thus the response is oblate at finite radius and becomes spherical as $R \rightarrow \infty$.

8 What is theorem vs assumption

- From the earlier Hysterical Response paper: $F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S})$ of ordinary gravity; no new force carrier.
- IR ballistic shell: local velocity isotropy + single IR scale $\Rightarrow S = J\delta(|u| - v)/(4\pi v^2)$.
- Ballistic spatial model of that residue: r^{-2} halo and quadrupole (continuum asymptotics tagged where Lean does not re-prove the integral expansion).
- Modelling closure: primary Poisson $\nabla^2\Phi_{\text{resp}} = \kappa n \Rightarrow g_{\text{resp}} = CQ/r$ (and the same map for q_{eff}); separate/unified as readings only.

- Transport PDE: retarded characteristic solution solves $v \cdot \nabla f = S - \Gamma f$ (Lean).
- ODE / attractor (theorem): given capacity inequality and filling dynamics, stable y_* and $Q^2 = (A^2/y_*)M_b$, hence BTFR exponent.
- Accretion $\rightarrow Q$ (Paper 1): $Q = \alpha J_{\text{in}}^{\text{tot}} = \alpha \dot{N}_{\text{acc}}$ when accretion supplies unentangled arrivals; not the unit identity $Q = \dot{M}_b$.
- Conjecture (not from the spatial model): formation/retention requires $\boxed{Q \gtrsim O(\sqrt{M_b})}$.
- Physical inputs (angular momentum): isotropic $j \propto M_b^{2/3}$ (homogeneous intergalactic dust/gas) and $R \sim j/v_f$ give $a = 5/12$.
- Not claimed: new force carriers; microscopic C, A, α ; Galactic numerical fits; proof of the formation requirement from the spatial model.